

6.7 Exercises

Exercise 6.1: Three looks at solving the LQ problem (LQP)

In the following exercise, you will write three codes to solve the LQR using Octave or MATLAB. The objective function is the LQR with mixed term

$$V = \frac{1}{2} \sum_{k=0}^{N-1} (x(k)' Q x(k) + u(k)' R u(k) + 2x(k)' M u(k)) + (1/2)x(N)' P_f x(N)$$

First, implement the method described in Section 6.1.1 in which you eliminate the state and solve the problem for the decision variable

$$\mathbf{u} = (u(0), u(1), \dots, u(N-1))$$

Second, implement the method described in Section 6.1.1 in which you do *not* eliminate the state and solve the problem for

$$\mathbf{z} = (u(0), x(1), u(1), x(2), \dots, u(N-1), x(N))$$

Third, use backward dynamic programming (DP) and the Riccati iteration to compute the closed-form solution for $u(k)$ and $x(k)$.

(a) Let

$$A = \begin{bmatrix} 4/3 & -2/3 \\ 1 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad C = \begin{bmatrix} -2/3 & 1 \end{bmatrix} \quad x(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$Q = C' C + 0.001 I \quad P_f = \Pi \quad R = 0.001 \quad M = 0$$

in which the terminal penalty, P_f is set equal to Π , the steady-state cost to go. Compare the three solutions for $N = 5$. Plot $x(k)$, $u(k)$ versus time for the closed-loop system.

(b) Let $N = 50$ and repeat. Do any of the methods experience numerical problems generating an accurate solution? Plot the condition number of the matrix that is inverted in the first two methods versus N .

(c) Now consider the following unstable system

$$A = \begin{bmatrix} 27.8 & -82.6 & 34.6 \\ 25.6 & -76.8 & 32.4 \\ 40.6 & -122.0 & 51.9 \end{bmatrix} \quad B = \begin{bmatrix} 0.527 & 0.548 \\ 0.613 & 0.530 \\ 1.06 & 0.828 \end{bmatrix} \quad x(0) = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Consider regulator tuning parameters and constraints

$$Q = I \quad P_f = \Pi \quad R = I \quad M = 0$$

Repeat parts (a) and (b) for this system. Do you lose accuracy in any of the solution methods? What happens to the condition number of $H(N)$ and $S(N)$ as N becomes large? Which methods are still accurate for this case? Can you explain what happened?

Exercise 6.2: LQ as least squares

Consider the standard LQP

$$\min_{\mathbf{u}} V = \frac{1}{2} \sum_{k=0}^{N-1} (x(k)' Q x(k) + u(k)' R u(k)) + (1/2) x(N)' P_f x(N)$$

subject to

$$x^+ = Ax + Bu$$

- Set up the dense Hessian least squares problem for the LQP with a horizon of three, $N = 3$. Eliminate the state equations and write out the objective function in terms of only the decision variables $u(0), u(1), u(2)$.
- What are the conditions for an optimum, i.e., what linear algebra problem do you solve to compute $u(0), u(1), u(2)$?

Exercise 6.3: Lagrange multiplier method

Consider the general least squares problem

$$\min_x V(x) = \frac{1}{2} x' H x + \text{const}$$

subject to

$$Dx = d$$

- What is the Lagrangian L for this problem? What is the dimension of the Lagrange multiplier vector, λ ?
- What are necessary and sufficient conditions for a solution to the optimization problem?
- Apply this approach to the LQP of Exercise 6.2 using the equality constraints to represent the model equations. What are H, D, d for the LQP?
- Write out the linear algebra problem to be solved for the optimum.
- Contrast the two different linear algebra problems in these two approaches. Which do you want to use when N is large and why?

Exercise 6.4: Reparameterizing an unstable system

Consider again the LQR problem with cross term

$$\min_{\mathbf{u}} V = \frac{1}{2} \sum_{k=0}^{N-1} (x(k)' Q x(k) + u(k)' R u(k) + 2x(k)' M u(k)) + (1/2) x(N)' P_f x(N)$$

subject to

$$x^+ = Ax + Bu$$

and the three approaches of Exercise 6.1.

- The method described in Section 6.1.1 in which you eliminate the state and solve the problem for the decision variable

$$\mathbf{u} = (u(0), u(1), \dots, u(N-1))$$

2. The method described in Section 6.1.1 in which you do *not* eliminate the state and solve the problem for

$$\mathbf{z} = (u(0), x(1), u(1), x(2), \dots, u(N-1), x(N))$$

3. The method of DP and the Riccati iteration to compute the closed-form solution for $u(k)$ and $x(k)$.

- (a) You found that unstable A causes numerical problems in the first method using large horizons. So let's consider a fourth method. Reparameterize the input in terms of a state feedback gain via

$$u(k) = Kx(k) + v(k)$$

in which K is chosen so that $A + BK$ is a stable matrix. Consider the matrices in a transformed LQP

$$\min_v V = \frac{1}{2} \sum_{k=0}^{N-1} \left(x(k)' \tilde{Q} x(k) + v(k)' \tilde{R} v(k) + 2x(k)' \tilde{M} v(k) \right) + (1/2) x(N)' \tilde{P}_f x(N)$$

subject to $x^+ = \tilde{A}x + \tilde{B}v$.

What are the matrices $\tilde{A}, \tilde{B}, \tilde{Q}, \tilde{P}_f, \tilde{R}, \tilde{M}$ such that the two problems give the same solution (state trajectory)?

- (b) Solve the following problem using the first method and the fourth method and describe differences between the two solutions. Compare your results to the DP approach. Plot $x(k)$ and $u(k)$ versus k .

$$A = \begin{bmatrix} 27.8 & -82.6 & 34.6 \\ 25.6 & -76.8 & 32.4 \\ 40.6 & -122.0 & 51.9 \end{bmatrix} \quad B = \begin{bmatrix} 0.527 & 0.548 \\ 0.613 & 0.530 \\ 1.06 & 0.828 \end{bmatrix} \quad x(0) = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Consider regulator tuning parameters and constraints

$$Q = P_f = I \quad R = I \quad M = 0 \quad N = 50$$

Exercise 6.5: Recursively summing quadratic functions

Consider generalizing Example 1.1 to an N -term sum. Let the N -term sum of quadratic functions be defined as

$$V(N, x) = \frac{1}{2} \sum_{i=1}^N (x - x(i))' X_i (x - x(i))$$

in which $x, x(i) \in \mathbb{R}^n$ are real n -vectors and $X_i \in \mathbb{R}^{n \times n}$ are positive definite matrices.

- (a) Show that $V(N, x)$ can be found recursively

$$V(N, x) = (1/2)(x - v(N))' H(N)(x - v(N)) + \text{constant}$$

in which $v(i)$ and $H(i)$ satisfy the recursion

$$H(i+1) = H_i + X_{i+1} \quad v(i+1) = H^{-1}(i+1) (H_i v_i + X_{i+1} x(i+1))$$

$$H_1 = X_1 \quad v_1 = x_1$$

Notice the recursively defined $v(m)$ and $H(m)$ provide the solutions and the Hessian matrices of the sequence of optimization problems

$$\min_x V(m, x) \quad 1 \leq m \leq N$$

- (b) Check your answer by solving the equivalent, but larger dimensional, constrained least squares problem (see Exercise 1.16)

$$\min_z (z - z_0)' \tilde{H} (z - z_0)$$

subject to

$$Dz = 0$$

in which $z, z_0 \in \mathbb{R}^{nN}$, $\tilde{H} \in \mathbb{R}^{nN \times nN}$ is a block diagonal matrix, $D \in \mathbb{R}^{n(N-1) \times nN}$

$$z_0 = \begin{bmatrix} x(1) \\ \vdots \\ x(N-1) \\ x(N) \end{bmatrix} \quad \tilde{H} = \begin{bmatrix} X_1 & & & \\ & \ddots & & \\ & & X_{N-1} & \\ & & & X_N \end{bmatrix} \quad D = \begin{bmatrix} I & -I & & \\ & \ddots & \ddots & \\ & & I & -I \end{bmatrix}$$

- (c) Compare the size and number of matrix inverses required for the two approaches.

Exercise 6.6: Why call the Lyapunov stability *nonuniform*?

Consider the following linear system

$$\begin{aligned} w^+ &= Aw & w(0) &= Hx(0) \\ x &= Cw \end{aligned}$$

with solution $w(k) = A^k w(0) = A^k Hx(0)$, $x(k) = CA^k Hx(0)$. Notice that $x(0)$ completely determines both $w(k)$ and $x(k)$, $k \geq 0$. Also note that zero is a solution, i.e., $x(k) = 0, k \geq 0$ satisfies the model.

- (a) Consider the following case

$$A = \rho \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \quad H = \begin{bmatrix} 0 \\ -1 \end{bmatrix} \quad C = \begin{bmatrix} 1 & -1 \end{bmatrix}$$

$$\rho = 0.925 \quad \theta = \pi/4 \quad x(0) = 1$$

Plot the solution $x(k)$. Does $x(k)$ converge to zero? Does $x(k)$ achieve zero exactly for finite $k > 0$?

- (b) Is the zero solution $x(k) = 0$ Lyapunov stable? State your definition of Lyapunov stability, and prove your answer. Discuss how your answer is consistent with the special case considered above.

Exercise 6.7: Exponential stability of suboptimal MPC with unbounded feasible set

Consider again Lemma 6.5 when both $\mathbb{U}$ and X_N are unbounded. Show that the suboptimal MPC controller is exponentially stable on the following sets.

- (a) Any sublevel set of $V(x, \mathbf{h}(x))$
- (b) Any compact subset of X_N

Exercise 6.8: A refinement to the warm start

Consider the following refinement to the warm start in the suboptimal MPC strategy. First add the requirement that the initialization strategy satisfies the following bound

$$\mathbf{h}(\mathbf{x}) \leq \tilde{d} \|\mathbf{x}\| \quad \mathbf{x} \in \mathcal{X}_N$$

in which $\tilde{d} > 0$. Notice that all initializations considered in the chapter satisfy this requirement.

Then, at time k and state $\mathbf{x}$, in addition to the shifted input sequence from time $k - 1$, $\tilde{\mathbf{u}}$, evaluate the initialization sequence applied to the current state, $\mathbf{u} = \mathbf{h}(\mathbf{x})$. Select whichever of these two input sequence has lower cost as the warm start for time k . Notice also that this refinement makes the constraint

$$\|\mathbf{u}\| \leq d \|\mathbf{x}\| \quad \mathbf{x} \in r\mathcal{B}$$

redundant, and it can be removed from the MPC optimization.

Prove that this refined suboptimal strategy is exponentially stabilizing on the set $\mathcal{X}_N$. Notice that with this refinement, we do not have to assume that $\mathcal{X}_N$ is bounded or that $\mathbb{U}$ is bounded.

Exercise 6.9: Global asymptotic stability and exponential convergence with mixed powers of the norm

Prove Lemma 6.6.

Hints: exponential convergence can be established as in standard exponential stability theorems. To establish Lyapunov stability, notice that $\|\mathbf{x}(0)\| \leq \|(x(0), e(0))\|$ and $\|e(0)\| \leq \|(x(0), e(0))\|$ and that $(\cdot)^\alpha$ for $\alpha > 0$ is a $\mathcal{K}_\infty$ function.

Exercise 6.10: Decentralized control of Examples 6.9–6.11

Apply decentralized control to the systems in Examples 6.9–6.11. Which of these systems are closed-loop unstable with decentralized control? Compare this result to the result for noncooperative MPC.

Exercise 6.11: Cooperative control of Examples 6.9–6.11

Apply cooperative MPC to the systems in Examples 6.9–6.11. Are any of these systems closed-loop unstable? Compare the closed-loop eigenvalues of converged cooperative control to centralized MPC, and discuss any differences.

Exercise 6.12: Adding norms

Establish the following result used in the proof of Lemma 6.14. Given that $\mathbf{w} \in \mathbb{R}^m$, $\mathbf{e} \in \mathbb{R}^n$

$$\frac{1}{\sqrt{2}}(\|\mathbf{w}\| + \|\mathbf{e}\|) \leq \|(\mathbf{w}, \mathbf{e})\| \leq \|\mathbf{w}\| + \|\mathbf{e}\| \quad \forall \mathbf{w}, \mathbf{e}$$

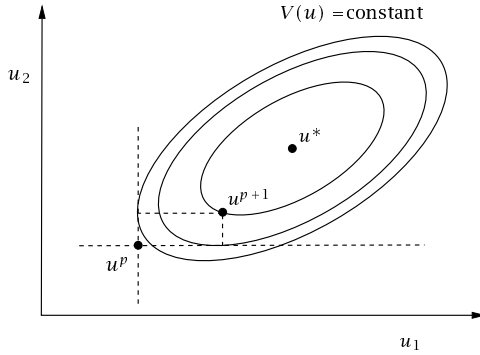


Figure 6.11: Optimizing a quadratic function in one set of variables at a time.

Exercise 6.13: Padding matrices

Given a vector $\mathbf{z}$ and subvector $\mathbf{u}$

$$\mathbf{z} = \begin{bmatrix} u(0) \\ x(1) \\ u(1) \\ x(2) \\ \vdots \\ u(N-1) \\ x(N) \end{bmatrix} \quad \mathbf{u} = \begin{bmatrix} u(0) \\ u(1) \\ \vdots \\ u(N-1) \end{bmatrix} \quad x \in \mathbb{R}^n \quad u \in \mathbb{R}^m$$

and quadratic function of $\mathbf{u}$

$$(1/2)\mathbf{u}'H\mathbf{u} + h'\mathbf{u}$$

Find the corresponding quadratic function of $\mathbf{z}$ so that

$$(1/2)\mathbf{z}'H_z\mathbf{z} + h'_z\mathbf{z} = (1/2)\mathbf{u}'H\mathbf{u} + h'\mathbf{u} \quad \forall \mathbf{z}, \mathbf{u}$$

Hint: first find the padding matrix E such that $\mathbf{u} = E\mathbf{z}$.

Exercise 6.14: A matrix inverse

Compute the four partitioned elements in the two-player feedback gain $(I - L)^{-1}\bar{K}$

$$\mathbf{u}^\infty = (I - L)^{-1}\bar{K}x(0) \quad |\text{eig}(L)| < 1$$

in which

$$(I - L)^{-1}\bar{K} = \begin{bmatrix} I & -L_1 \\ -L_2 & I \end{bmatrix}^{-1} \begin{bmatrix} K_1 & 0 \\ 0 & K_2 \end{bmatrix}$$

Exercise 6.15: Optimizing one variable at a time

Consider the positive definite quadratic function partitioned into two sets of variables

$$V(u) = (1/2)u'Hu + c'u + d$$

$$V(u_1, u_2) = (1/2) \begin{bmatrix} u_1' & u_2' \end{bmatrix} \begin{bmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} + \begin{bmatrix} c_1' & c_2' \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} + d$$

in which $H > 0$. Imagine we wish to optimize this function by first optimizing over the u_1 variables holding u_2 fixed and then optimizing over the u_2 variables holding u_1 fixed as shown in Figure 6.11. Let's see if this procedure, while not necessarily efficient, is guaranteed to converge to the optimum.

- (a) Given an initial point (u_1^p, u_2^p) , show that the next iteration is

$$u_1^{p+1} = -H_{11}^{-1} (H_{12}u_2^p + c_1)$$

$$u_2^{p+1} = -H_{22}^{-1} (H_{21}u_1^p + c_2) \quad (6.39)$$

The procedure can be summarized as

$$u^{p+1} = Au^p + b \quad (6.40)$$

in which the iteration matrix A and constant b are given by

$$A = \begin{bmatrix} 0 & -H_{11}^{-1}H_{12} \\ -H_{22}^{-1}H_{21} & 0 \end{bmatrix} \quad b = \begin{bmatrix} -H_{11}^{-1}c_1 \\ -H_{22}^{-1}c_2 \end{bmatrix} \quad (6.41)$$

- (b) Establish that the optimization procedure converges by showing the iteration matrix is stable

$$|\text{eig}(A)| < 1$$

- (c) Given that the iteration converges, show that it produces the same solution as

$$u^* = -H^{-1}c$$

Exercise 6.16: Monotonically decreasing cost

Consider again the iteration defined in Exercise 6.15.

- (a) Prove that the cost function is monotonically decreasing when optimizing one variable at a time

$$V(u^{p+1}) < V(u^p) \quad \forall u^p \neq -H^{-1}c$$

- (b) Show that the following expression gives the size of the decrease

$$V(u^{p+1}) - V(u^p) = -(1/2)(u^p - u^*)'P(u^p - u^*)$$

in which

$$P = HD^{-1}\tilde{H}D^{-1}H \quad \tilde{H} = D - N \quad D = \begin{bmatrix} H_{11} & 0 \\ 0 & H_{22} \end{bmatrix} \quad N = \begin{bmatrix} 0 & H_{12} \\ H_{21} & 0 \end{bmatrix}$$

and $u^* = -H^{-1}c$ is the optimum.

Hint: to simplify the algebra, first change coordinates and move the origin of the coordinate system to u^* .

Exercise 6.17: One variable at a time with convex step

Consider Exercise 6.15 but with the convex step for the iteration

$$\begin{bmatrix} u_1^{p+1} \\ u_2^{p+1} \end{bmatrix} = w_1 \begin{bmatrix} u_1^0(u_2^p) \\ u_2^p \end{bmatrix} + w_2 \begin{bmatrix} u_1^p \\ u_2^0(u_1^p) \end{bmatrix} \quad 0 \leq w_1, w_2 \quad w_1 + w_2 = 1$$

- (a) Show that the iteration for the convex step is also of the form

$$u^{p+1} = Au^p + b$$

and the A matrix and b vector for this case are

$$A = \begin{bmatrix} w_2 I & -w_1 H_{11}^{-1} H_{12} \\ -w_2 H_{22}^{-1} H_{21} & w_1 I \end{bmatrix} \quad b = \begin{bmatrix} -w_1 H_{11}^{-1} \\ -w_2 H_{22}^{-1} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

- (b) Show that A is stable.

- (c) Show that this iteration also converges to $u^* = -H^{-1}c$.

Exercise 6.18: Monotonically decreasing cost with convex step

Consider again the problem of optimizing one variable at a time with the convex step given in Exercise 6.17.

- (a) Prove that the cost function is monotonically decreasing

$$V(u^{p+1}) < V(u^p) \quad \forall u^p \neq -H^{-1}c$$

- (b) Show that the following expression gives the size of the decrease

$$V(u^{p+1}) - V(u^p) = -(1/2)(u^p - u^*)' P (u^p - u^*)$$

in which

$$P = HD^{-1}\tilde{H}D^{-1}H \quad \tilde{H} = D - N$$

$$D = \begin{bmatrix} w_1^{-1}H_{11} & 0 \\ 0 & w_2^{-1}H_{22} \end{bmatrix} \quad N = \begin{bmatrix} -w_1^{-1}w_2H_{11} & H_{12} \\ H_{21} & -w_1w_2^{-1}H_{22} \end{bmatrix}$$

and $u^* = -H^{-1}c$ is the optimum.

Hint: to simplify the algebra, first change coordinates and move the origin of the coordinate system to u^* .

Exercise 6.19: Splitting more than once

Consider the generalization of Exercise 6.15 in which we repeatedly decompose a problem into one-variable-at-a-time optimizations. For a three-variable problem we have the three optimizations

$$u_1^{p+1} = \arg \min_{u_1} V(u_1, u_2^p, u_3^p)$$

$$u_2^{p+1} = \arg \min_{u_2} V(u_1^p, u_2, u_3^p) \quad u_3^{p+1} = \arg \min_{u_3} V(u_1^p, u_2^p, u_3)$$

Is it true that

$$V(u_1^{p+1}, u_2^{p+1}, u_3^{p+1}) \leq V(u_1^p, u_2^p, u_3^p)$$

Hint: you may wish to consider the following example, $V(u) = (1/2)u'Hu + c'u$, in which

$$H = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix} \quad c = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \quad u^p = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Exercise 6.20: Time-varying controller iterations

We let $p_k \geq 0$ be a time-varying integer-valued index representing the iterations applied in the controller at time k .

$$\begin{aligned}x_1(k+1) &= A_1 x_1(k) + \bar{B}_{11} u_1(0;k) + \bar{B}_{12} u_2(0;k) \\x_2(k+1) &= A_2 x_2(k) + \bar{B}_{21} u_1(0;k) + \bar{B}_{22} u_2(0;k) \\u_1(k+1) &= g_1^{p_k}(x_1(k), x_2(k), u_1(k), u_2(k)) \\u_2(k+1) &= g_2^{p_k}(x_1(k), x_2(k), u_1(k), u_2(k))\end{aligned}$$

Notice the system evolution is time-varying even though the models are time invariant because we allow a time-varying sequence of controller iterations.

Show that cooperative MPC is exponentially stabilizing for any $p_k \geq 0$ sequence.

Exercise 6.21: Stable interaction models

In some industrial applications it is preferable to partition the plant so that there are no unstable connections between subsystems. Any inputs u_j that have unstable connections to outputs y_i should be included in the i th subsystem inputs. Allowing an unstable connection between two subsystems may not be robust to faults and other kinds of system failures.⁵ To implement this design idea in the two-player case, we replace Assumption 6.13 (b) with the following

Modified Assumption 6.13 (Constrained two-player game).

(b) The interaction models A_{ij} , $i \neq j$ are stable.

Prove that Modified Assumption 6.13 (b) implies Assumption 6.13 (b). It may be helpful to first prove the following lemma.

Lemma 6.23 (Local detectability). *Given partitioned system matrices*

$$A = \begin{bmatrix} A & 0 \\ 0 & A_s \end{bmatrix} \quad C = \begin{bmatrix} C & C_s \end{bmatrix}$$

in which A_s is stable, the system (A, C) is detectable if and only if the system (A, C) is detectable.

Hint: use the Hautus lemma as the test for detectability.

Next show that this lemma and Modified Assumption 6.13 (b) establishes the distributed detectability assumption, Assumption 6.13 (b).

Exercise 6.22: Norm constraints as linear inequalities

Consider the quadratic program (QP) in decision variable u with parameter x

$$\begin{aligned}\min_u \quad & (1/2)u' H u + x' D u \\ \text{s.t.} \quad & E u \leq F x\end{aligned}$$

⁵We are not considering the common instability of base-level inventory management in this discussion. It is assumed that level control in storage tanks (integrators) is maintained at all times with simple, local level controllers. The internal unit flowrates dedicated for inventory management are not considered available inputs in the MPC problem.

in which $u \in \mathbb{R}^m$, $x \in \mathbb{R}^n$, and $H > 0$. The parameter x appears linearly (affinely) in the cost function and constraints. Assume that we wish to add a norm constraint of the following form

$$\|u\|_\alpha \leq c \|x\|_\alpha \quad \alpha = 2, \infty$$

- (a) If we use the infinity norm, show that this problem can be posed as an equivalent QP with additional decision variables, and the cost function and constraints remain linear (affine) in parameter x . How many decision variables and constraints are added to the problem?
- (b) If we use the two norm, show that this problem can be approximated by a QP whose solution does satisfy the constraints, but the solution may be suboptimal compared to the original problem.

Exercise 6.23: Steady-state noncooperative game

Consider again the steady-state target problem for the system given in Example 6.12.

- (a) Resolve the problem for the choice of convex step parameters $w_1 = 0.2$, $w_2 = 0.8$. Does the iteration for noncooperative control converge? Plot the iterations for the noncooperative and cooperative cases.
- (b) Repeat for the convex step $w_1 = 0.8$, $w_2 = 0.2$. Are the results identical to the previous part? If not, discuss any differences.
- (c) For what choices of w_1, w_2 does the target iteration converge using noncooperative control for the target calculation?

Exercise 6.24: Optimality conditions for constrained optimization

Consider the convex quadratic optimization problem

$$\min_u V(u) \quad \text{subject to} \quad u \in \mathbb{U}$$

in which V is a convex quadratic function and $\mathbb{U}$ is a convex set. Show that u^* is an optimal solution if and only if

$$\langle z - u^*, -\nabla V|_{u^*} \rangle \leq 0 \quad \forall z \in \mathbb{U} \quad (6.42)$$

Figure 6.12(a) depicts this condition for $u \in \mathbb{R}^2$. This condition motivates defining the normal cone (Rockafellar, 1970) to $\mathbb{U}$ at u^* as follows

$$N(\mathbb{U}, u^*) = \{y \mid \langle z - u^*, y - u^* \rangle \leq 0 \quad \forall z \in \mathbb{U}\}$$

The optimality condition can be stated equivalently as u^* is an optimal point if and only if the negative gradient is in the normal cone to $\mathbb{U}$ at u^*

$$-\nabla V|_{u^*} \in N(\mathbb{U}, u^*)$$

This condition and the normal cone are depicted in Figure 6.12(b).

Exercise 6.25: Partitioned optimality conditions with constraints

Consider a partitioned version of the constrained optimization problem of Exercise 6.24 with uncoupled constraints

$$\min_{u_1, u_2} V(u_1, u_2) \quad \text{subject to} \quad u_1 \in \mathbb{U}_1 \quad u_2 \in \mathbb{U}_2$$

in which V is a quadratic function and $\mathbb{U}_1$ and $\mathbb{U}_2$ are convex and nonempty.

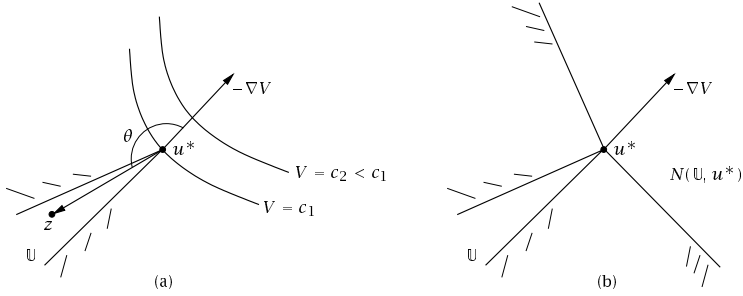


Figure 6.12: (a) Optimality of u^* means the angle between $-\nabla V$ and any point z in the feasible region must be greater than 90° and less than 270° . (b) The same result restated: u^* is optimal if and only if the negative gradient is in the normal cone to the feasible region at u^* , $-\nabla V|_{u^*} \in N(\mathbb{U}, u^*)$.

(a) Show that (u_1^*, u_2^*) is an optimal solution if and only if

$$\begin{aligned} \langle z_1 - u_1^*, -\nabla_{u_1} V|_{(u_1^*, u_2^*)} \rangle &\leq 0 \quad \forall z_1 \in \mathbb{U}_1 \\ \langle z_2 - u_2^*, -\nabla_{u_2} V|_{(u_1^*, u_2^*)} \rangle &\leq 0 \quad \forall z_2 \in \mathbb{U}_2 \end{aligned} \quad (6.43)$$

(b) Extend the optimality conditions to cover the case

$$\min_{u_1, \dots, u_M} V(u_1, \dots, u_M) \quad \text{subject to} \quad u_j \in \mathbb{U}_j \quad j = 1, \dots, M$$

in which V is a quadratic function and the $\mathbb{U}_j$ are convex and nonempty.

Exercise 6.26: Constrained optimization of M variables

Consider an optimization problem with M variables and uncoupled constraints

$$\min_{u_1, \dots, u_M} V(u_1, u_2, \dots, u_M) \quad \text{subject to} \quad u_l \in \mathbb{U}_j \quad j = 1, \dots, M$$

in which V is a strictly convex function. Assume that the feasible region is convex and nonempty and denote the unique optimal solution as $(u_1^*, u_2^*, \dots, u_M^*)$ having cost $V^* = V(u_1^*, \dots, u_M^*)$. Denote the M one-variable-at-a-time optimization problems at iteration k

$$z_j^{p+1} = \arg \min_{u_j} V(u_1^p, \dots, u_j, \dots, u_M^p) \quad \text{subject to} \quad u_j \in \mathbb{U}_j$$

Then define the next iterate to be the following convex combination of the previous and new points

$$\begin{aligned} u_j^{p+1} &= \alpha_j^p z_j^{p+1} + (1 - \alpha_j^p) u_j^p \quad j = 1, \dots, M \\ \varepsilon &\leq \alpha_j^p < 1 \quad 0 < \varepsilon \quad j = 1, \dots, M, \quad p \geq 1 \end{aligned}$$

$$\sum_{j=1}^M \alpha_j^p = 1, \quad p \geq 1$$

Prove the following results.

- (a) Starting with any feasible point, $(u_1^0, u_2^0, \dots, u_M^0)$, the iterations $(u_1^p, u_2^p, \dots, u_M^p)$ are feasible for $p \geq 1$.
- (b) The objective function decreases monotonically from any feasible initial point
- $$V(u_1^{p+1}, \dots, u_M^{p+1}) \leq V(u_1^p, \dots, u_M^p) \quad \forall u_l^0 \in \mathbb{U}_j, j = 1, \dots, M, \quad p \geq 1$$
- (c) The cost sequence $V(u_1^p, u_2^p, \dots, u_M^p)$ converges to the optimal cost V^* from any feasible initial point.
- (d) The sequence $(u_1^p, u_2^p, \dots, u_M^p)$ converges to the optimal solution $(u_1^*, u_2^*, \dots, u_M^*)$ from any feasible initial point.

Exercise 6.27: The constrained two-variable special case

Consider the special case of Exercise 6.26 with $M = 2$

$$\min_{u_1, u_2} V(u_1, u_2) \quad \text{subject to} \quad u_1 \in \mathbb{U}_1 \quad u_2 \in \mathbb{U}_2$$

in which V is a strictly positive quadratic function. Assume that the feasible region is convex and nonempty and denote the unique optimal solution as (u_1^*, u_2^*) having cost $V^* = V(u_1^*, u_2^*)$. Consider the two one-variable-at-a-time optimization problems at iteration k

$$\begin{aligned} u_1^{p+1} &= \arg \min_{u_1} V(u_1, u_2^p) & u_2^{p+1} &= \arg \min_{u_2} V(u_1^p, u_2) \\ \text{subject to } u_1 &\in \mathbb{U}_1 & \text{subject to } u_2 &\in \mathbb{U}_2 \end{aligned}$$

We know from Exercise 6.15 that taking the full step in the unconstrained problem with $M = 2$ achieves a cost decrease. We know from Exercise 6.19 that taking the full step for an unconstrained problem with $M \geq 3$ does *not* provide a cost decrease in general. We know from Exercise 6.26 that taking a reduced step in the constrained problem for all M achieves a cost decrease. That leaves open the case of a full step for a constrained problem with $M = 2$.

Does the full step in the constrained case for $M = 2$ guarantee a cost decrease? If so, prove it. If not, provide a counterexample.

Exercise 6.28: Subsystem stability constraints

Show that the following uncoupled subsystem constraints imply an overall system constraint of the same type. The first is suitable for asymptotic stability and the second for exponential stability.

- (a) Given $r_1, r_2 > 0$, and functions γ_1 and γ_2 of class $\mathcal{K}$, assume the following constraints are satisfied

$$\begin{aligned} |\mathbf{u}_1| &\leq \gamma_1(|x_1|) \quad x_1 \in r_1 \mathcal{B} \\ |\mathbf{u}_2| &\leq \gamma_2(|x_2|) \quad x_2 \in r_2 \mathcal{B} \end{aligned}$$

Show that there exists $r > 0$ and function γ of class $\mathcal{K}$ such that

$$|(\mathbf{u}_1, \mathbf{u}_2)| \leq \gamma(|(x_1, x_2)|) \quad (x_1, x_2) \in r \mathcal{B} \quad (6.44)$$

- (b) Given $r_1, r_2 > 0$, and constants $c_1, c_2, \sigma_1, \sigma_2 > 0$, assume the following constraints are satisfied

$$\begin{aligned} |\mathbf{u}_1| &\leq c_1 |x_1|^{\sigma_1} & x_1 &\in r_1 \mathcal{B} \\ |\mathbf{u}_2| &\leq c_2 |x_2|^{\sigma_2} & x_2 &\in r_2 \mathcal{B} \end{aligned}$$

Show that there exists $r > 0$ and function $c, \sigma > 0$ such that

$$|(\mathbf{u}_1, \mathbf{u}_2)| \leq c |(x_1, x_2)|^\sigma \quad (x_1, x_2) \in r\mathcal{B} \quad (6.45)$$

Exercise 6.29: Distributed disturbance detectability

Prove Lemma 6.16.

Hint: use the Hautus lemma as the test for detectability.

Exercise 6.30: Distributed target problem and uncoupled constraints

Player one's distributed target problem in the two-player game is given in (6.28)

$$\min_{x_{11s}, x_{21s}, u_{1s}} (1/2) \begin{bmatrix} H_1 y_{1s} - z_{1sp} \\ H_2 y_{2s} - z_{2sp} \end{bmatrix}' \begin{bmatrix} T_{1s} & \\ & T_{2s} \end{bmatrix} \begin{bmatrix} H_1 y_{1s} - z_{1sp} \\ H_2 y_{2s} - z_{2sp} \end{bmatrix}$$

subject to

$$\begin{bmatrix} I - A_1 & & -\bar{B}_{11} & -\bar{B}_{12} \\ & I - A_2 & -\bar{B}_{21} & -\bar{B}_{22} \end{bmatrix} \begin{bmatrix} x_{1s} \\ x_{2s} \\ u_{1s} \\ u_{2s} \end{bmatrix} = \begin{bmatrix} B_{1d} \hat{d}_1(k) \\ B_{2d} \hat{d}_2(k) \end{bmatrix}$$

$$E_1 u_{1s} \leq e_1$$

Show that the constraints can be expressed so that the target problem constraints are uncoupled.

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